RIPHAH International Colleges

A Project of Riphah International University

Course Outline

Course Information	Course Title	**Object Oriented Programming**		
	Course ID	CS-2104/ CS-1154	Course Type	Computing Core
	Credit hours	4 (3-3)	Hours per week	3-3
	Programs	ADP Computing (CS, AI, DS) ADP Computer System	Preferred Semester	2nd
Course Description	This course is designed as an entry level programming course for students who have prior programming experience. This course introduces the concepts of object-oriented programming to students with a background in the procedural paradigm. The course begins with a brief review structured data type. It then moves on to introduce the object-oriented programming paradigm, focusing on the definition and use of classes along with the fundamentals of object-oriented design. Other topics include an overview of programming language principles, simple analysis of algorithms, and an introduction to software engineering issues. Brief review of control structures, functions, and primitive data types Object-oriented programming: Object-oriented design; encapsulation and information-hiding; separation of behavior and implementation; classes, subclasses, and inheritance; polymorphism; class hierarchies; dynamic memory allocation, File handling			
Course Objectives	The objective of this course is to enable students;			
	No.	Objective		Relation with Program Objectives
	1.	At the end of the class, we expect people to have a good understanding about the concept of object-oriented programming using C++, be able to write and read basic C++ code.		PLO 1,
	2.	Using inheritance and polymorphism to manage the source code.		PLO 4, 5
	3.	Student will be motivated and encouraged to choose a practical project which should be management system or transactional system.		PLO 1-6
Course Learning Outcomes (CLO)	At the end of this course students will be able to demonstrate;			
	No.	Outcome		Relation with SLO/PLO
	1.	Understand the relative merits of C++ as an object oriented		PLO 1,

		programming language				
	2.	Understand the features of C++ supporting object-oriented program and produce object-oriented software using C++				PLO 4, 5, 6, 10, 11
	3.	Understand how to apply the major object-oriented concepts to implement object-oriented programs in C++, encapsulation, inheritance and polymorphism				PLO 8, 12
	4.	Understand advanced features of C++ specifically stream I/O, templates and operator overloading				PLO 6, 7, 9
Lecture type	Class room Lectures, Lab Sessions, Project Presentation					
Prerequisites	Programming Fundamental I					
Follow up Courses	Data Structures					
Course Software or Tool	Visual Studio, Dev C++					
Textbook	Title	Edition	Authors	Publisher	Year	ISBN
	Object Oriented Programming	4th	Robert Lafore	Sams	December 2001	0-672-32308-7
References	C++ Programming: Program Design Including Data Structures	6th	D.S. Malik			978-1-13362628-1
Assessment Criteria (100%)	Assessment	Weight			Used to attain CLO	
	Assignments & Quiz	10%			CLO1-CLO4	
	Project & Presentation	10%			CLO1-CLO4	
	Lab	15%			CLO1-CLO4	
	Mid Term	25%			CLO1-CLO4	
	Final	40%			CLO1-CLO4	
Methods of Evaluation	Assignments, Quizzes, Project, Midterm paper, Final term paper					

Week No.	Topic	Lecture No.	Lecture Contents	Relation with CLO	Task
W1.	Introduction to OOP	L1.	• Module Discussion • Introduction to OOP • Primitive Data types and User defined datatypes • Structure • Declaration of a simple structure • Defining a structure variable • Accessing members of the structure • Initialization of a structure variables	CLO 1	C++ Object Basics: Functions, Recursion, and Objects Read: Week 1 Notes
		L2.	• Nested Structure • Declaration, Definition, Accessing members and initialization of nested structure • Enumerations	CLO 1	
W2.	Objects and Classes	L3.	• Characteristics of Object-Oriented Languages • Defining the Class • Using the Class • Calling Member Functions	CLO 2	Read: Week 3 Notes
		L4.	• Examples of class • Member Functions Defined Outside the Class	CLO 2	Assignment: C++ Classes and Objects Coursera Project
W3.	Constructors	L5.	• Objects as Function Arguments • Objects as Arguments	CLO 2	Read: Week 4 Notes
		L6.	• Defining Constructor • Overloaded Constructors • The Default Copy Constructor	CLO 2	
W4.	Static Class Data	L7.	• Defining Destructors • Example of Destructors	CLO 2	
		L8.	• Uses of Static Class Data • An Example of Static Class Data	CLO 2	
W5.	Inheritance	L9.	• Introduction to inheritance • Derived Class and Base Class	CLO 2, 3	
		L10.	• Accessing Base Class Members • The protected Access Specifier	CLO 2, 3	Object-Oriented C++: Inheritance and Encapsulation Read: Week 1 notes
W6.	Type of Inheritance	L11.	• Derived Class Constructors • Base Class Constructors	CLO 2, 3	
		L12.	• Public Inheritance • Protected Inheritance • Private Inheritance	CLO 2, 3	Read: Week 2 Notes 1, 2
W7.		L13.	• Levels of Inheritance	CLO 2, 3	

	Levels of Inheritance	L14.	• Multiple Inheritance • Member Functions in Multiple Inheritance	CLO 2, 3	Read: Week 2 Notes 3
W8.	Polymorphism	L15.	• Polymorphism • Advantages of Using Polymorphism		
		L16.	• Pointers • Pointer to Objects		
W9.	Mid Term Exam Week	L17.	**Mid Term Examination**		
		L18.			
W10.	Polymorphism	L19.	• Pointer to Objects with coding example • Early binding concept	CLO 3	Read: Week 3 Notes
		L20.	• Polymorphism with Virtual Function Concept • Implementation	CLO 3	
W11.	Virtual Functions	L21.	• Virtual Function • Abstract Base Class & Concrete Derived Class	CLO 3	Assignment: C++ Inheritance, Aggregation and Composition
		L22.	• Normal Member Functions Accessed with • Pointers • Virtual Member Functions Accessed with Pointers	CLO 3	
W12.	Late Binding & Friend Functions	L23.	• Abstract Classes and Pure Virtual • Functions • Virtual Destructors	CLO 3	
		L24.	• Virtual Base Classes • Friend Function • Friend Class	CLO 2, 3	
W13.	Static Functions	L25.	• Static Functions • Accessing static Functions	CLO 2, 3	
	Unary Operators	L26.	• The operator Keyword • Overloading Unary Operators	CLO 2, 3	
W14.		L27.	• Operator Arguments • Operator Return Values	CLO 4	
	Overloading Binary Operators & File Handling	L28.	• Arithmetic Operators, Concatenating Strings • Multiple Overloading • Comparison Operators,	CLO 4	
W15.		L29.	• Arithmetic Assignment Operators • File Handling	CLO 4	
		L30.	• Writing Data, Reading Data • Appending files, Deleting Records	CLO 4	
W16.	Function Templates	L31.	• Function Templates, A Simple Function Template, Function Template Syntax, What the Compiler Does, The Deciding Argument. Template Arguments Must Match, Why Not Macros? Class Templates.	CLO 4	
		L32.	• Exception Handling. Try block, catch block and throw statement. Multiple Exception Handling		
W17	Project Presentation	L33.	• Project Presentation	CLO 1-4	
		L34.	• Project Presentation		
W18.	**Final Exam**				

<u>Object Oriented Programming Lab</u>

Week #	Topic	Details	Homework
1.	Introduction to the Course and C++ Programming	• Course Outline • Assessment Policies • History of Programming Languages • Features of C++ • Overview of basic C++ concepts • Required tools and their configurations • Edit, compile, and run C++ application	None
2.	Structures	• Declaration of a simple structure • Defining a structure variable • Nested Structures	Textbook Exercise
3.	Introduction to Object Oriented Paradigm	• Classes and Objects • Object Oriented vs Structured Programming • Member functions and Data members	
4.	Object Oriented Analysis & Design	• OOAD • Use of UML for OOAD	Online Exercise
5.	Classes and Objects in C++	• C++ Classes & Objects • Object Instantiation • Instance Variables • Class Variables • Constructors • Instance Methods • Class Methods • The this keyword • Passing and returning objects	Textbook Exercise
6.	Classes and Objects in C++	• Controlling Access to Members • Set and Get Methods • static keyword usage • const keyword usage	
7.	Object Oriented Programming with C++	• Abstraction • Inheritance • Method Overriding • Method Overloading • Composition • Associations • Delegation	Textbook Exercise
8	Polymorphism using pointers	• Pointers • Pointer to Objects • Examples	Textbook Exercise
9.	Mid Term Week	• Mid Term Examination	
10.	Polymorphism using Abstract Classes	• Polymorphism • Abstract Class • Abstract class inheritance	Textbook Exercise
11.	Methods & Arrays: A Deeper Look	• static Methods, static Fields • Method Signatures • Declaring Methods with Multiple Parameters • Passing Objects to Methods • Method-Call Stack and Stack Frames • Argument Promotion and Casting • Scope of Declarations • Passing Arrays to Methods • Pass-By-Value vs. Pass-By-Reference • Arrays of Objects • Multidimensional Arrays	Textbook Exercise

Week #	Topic	Details	Homework
12.	File Processing	• Files and streams • Creating a sequential file • Reading and updating a sequential file • Random access files	Textbook Exercise
13.	Exception Handling	• Why Exceptions • Standard Exception Handling Options • Exception Class Hierarchy • Catching an Exception: try and catch blocks • Rethrowing exception	Textbook Exercise
14.	Exception Handling	• Methods Which Throw Exceptions: the throws clause • Stack unwinding • Handling new in exception handling • Use of unique pointer • Writing Custom Exceptions	Textbook Exercise
15.	Templates and Operator Overloading	• Class Templates • Writing your own template classes • Operator overloading	Textbook Exercise
16.	Function Template	• A Simple Function Template, Function Template Syntax, What the Compiler Does, The Deciding Argument. Template Arguments Must Match, Why Not Macros? Class Templates.	Textbook Exercise
17.	Project Demonstrations	• Development Project presentations	
18.	Final exam	• Final exam	None